11-18 第2篇

阅读理解（说明文） Why Does the World Feel So Quiet When It Snows?

① Every year on the first heavy snowfall of winter, something strange happens to my neighbourhood. At 06:45, I lift the curtain, and the street — the same street where buses usually shake the window and delivery cars beep impatiently — has turned into a **quiet photograph**. I can hear my own breath. I can hear the low refrigerator in the kitchen through two closed doors. A delivery man walks past, dragging a large red parcel, and the sound of his footsteps on the white ground is softer than a whisper. People often describe it in the same way: "When it snows, the world feels like someone has pressed a **soft mute button** on everything." For years I thought this was only a feeling. In fact, it is a **real physical effect** — and a very beautiful one.

② To understand the quiet of snow, we first need to understand what sound is. When someone shouts, or a car beeps, the air around them **shakes** back and forth in tiny waves. Those waves travel through the air, reach our ears, and wiggle our eardrums. The louder the sound, the bigger the waves. But what happens when those air waves hit a **soft, hollow, many-holed material** on the ground?

③ The answer is: the waves get **trapped**. Fresh snow is not a solid sheet of ice. Every snowflake is a tiny, six-armed star with thousands of small **open spaces between its arms**. When many fresh snowflakes land, they do not pack closely. They leave a large number of **tiny air pockets** between them — so much so that a 10-centimetre layer of freshly fallen snow is, by volume, **more than 90% air**. The result is a surface that looks soft, feels fluffy, and, above all, **swallows sound**.

④ When a sound wave enters one of those tiny pockets, the air inside shakes; some of the shaking energy is **turned into heat** through tiny rubs between the air and the snow. The wave that eventually comes back out is **much weaker**. Scientists measure this effect with a number called the **absorption coefficient**. For a typical hard road, it is about **0.02** — only 2% of sound is swallowed. For fresh snow, it can be as high as **0.8—0.9** — up to 90% of sound is gone. This is why, on a snowy morning, even a large city can feel like a library.

⑤ But there is a second reason, just as interesting: **the air above the snow also changes**. On a cold, snowy day, the ground is very cold and the air just above it is cold; the air higher up (a few hundred metres up) may still be **a little warmer**. This unusual situation makes sound waves **bend gently upwards**, away from the ground. A car horn blown three streets away, which on a normal day would reach your ear through the bus

stop, now curves up and **disappears into the upper sky**. Even sound that is not directly absorbed by the snow is being, quietly, redirected away from you.

⑥ A third reason lies in **our own behaviour**. A heavy snowfall keeps fewer people outside, and those who do go out walk more carefully. Fewer feet on the ground, fewer engines running, fewer birds calling. The city **makes less sound in the first place**. On top of that, snowflakes slow down **small fast wind**, which is usually a big source of the background whisper in the air. The world, for a short while, simply agrees to be quieter.

⑦ These three effects — absorption, upward bending, and lower human noise — together create that rare calm. It does not last forever. After one or two days, the top of the snow is **melted slightly by the sun**, then freezes again at night, forming a thin hard crust. On this hard crust, tiny air pockets disappear. The absorption drops back towards 0.2; the world begins to return to its usual noise. The unique stillness of "fresh snow morning" is short-lived — which may be why we remember it so clearly.

⑧ So the next time you open the curtain on the first white morning, do not only look at the snow; **listen to it**. Step carefully onto the front step, stand still for half a minute, and let your ears notice what is no longer there: the heavy buses, the loud horns, the distant whistle of the train. In their place, you may hear things you had forgotten existed — the very small click of your own coat button, the low sound of the wind passing through a bamboo leaf, a single bird singing three blocks away. The winter, by taking away some of the loud world, hands us back, for one morning, **a little piece of our own hearing**. That is a gift worth stepping out into the cold to receive.

注释： mute button 静音键；eardrum 鼓膜；absorption 吸收；coefficient 系数；crust 硬壳

根据短文内容，选择最佳答案。

1. What is the first reason snow makes the world quieter?

A. Snowflakes produce a faint soothing sound.
B. Fresh snow is mostly air: a 10-cm layer of fresh snow is more than 90% air, full of tiny pockets between the arms of the snowflakes; when sound waves enter, the air inside them shakes and some shaking energy is turned into heat by tiny rubs, so the waves come back out much weaker — fresh snow can absorb up to 80—90% of sound.
C. Snow freezes sound.
D. Snow is too heavy to move.

2. What role does the COLD LAYER of air near the ground play in quieting the city?

A. It keeps the snow white.
B. It makes cars run more slowly.
C. The ground is very cold and the air just above it is cold, while the air a few hundred metres higher may be a little warmer; this makes sound waves bend gently upwards,

away from the ground, so sounds that would normally reach the ear curve up and disappear into the upper sky.
D. It makes snowflakes grow bigger.

3. What do people DO less on a snowy morning, according to Paragraph ⑥?

A. They sleep more but talk more.
B. They walk more carefully, fewer feet are on the ground, fewer engines run, and fewer birds call; the city simply makes less sound in the first place.
C. They play louder music.
D. They buy more cars.

4. Why doesn't the special quiet of fresh snow last forever?

A. Because the snow crystals always grow bigger.
B. After one or two days, the top of the snow is slightly melted by the sun and freezes again at night, forming a thin hard crust; on this crust, tiny air pockets disappear, and the absorption drops back towards 0.2; the world begins to return to its usual noise.
C. Because more buses suddenly arrive.
D. Because snow turns into rain.

5. What is the main message of the last paragraph?

A. Snow is dangerous and should be avoided.
B. The next time you open the curtain on the first white morning, do not only look at the snow — listen to it; the winter, by taking away some of the loud world, hands us back a little piece of our own hearing, a gift worth stepping out into the cold to receive.
C. Snow makes everyone deaf.
D. Students should not listen to snow.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第③、④段吸音机制
【解析】新雪吸声直引。A、C、D 均错。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第⑤段 "cold ground ... warmer air above ... sound bends upwards"
【解析】反温层折射直引。A、B、D 均错。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第⑥段人类行为部分
【解析】人与环境原因直引。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑦段硬壳成因
【解析】表层硬化后吸音下降直引。A、C、D 均错。故选 B。

5.
【答案】B
【题型】推理判断题
【定位】第⑧段结尾
【解析】"冬雪归还听觉"的主题。A、C、D 均偏离。故选 B。